
PS Analysis in Data Science: Problem Set 6

Problem 1. (... used in the proof of the Taylor formula)

- (a) Suppose that $\alpha \in \mathbb{N}^n$ and $h \in \mathbb{R}^n$ such that $\|h\|_\infty \leq 1$. Prove that

$$|h^\alpha| \leq \|h\|^{|\alpha|}.$$

- (b) Let $k \in \mathbb{N}$ and suppose that $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is k times continuously differentiable. For $x, x_0 \in \mathbb{R}^n$ and $h = x - x_0$, define

$$g : [-1, 1] \rightarrow \mathbb{R}, \quad g(t) := f(x_0 + th).$$

For $j \in \{0, \dots, k\}$ prove that g is k -times continuously differentiable and

$$g^{(j)}(t) = \sum_{\substack{\alpha \in \mathbb{N}^n \\ |\alpha|=j}} \frac{j!}{\alpha!} (\partial^\alpha f)(x_0 + th) h^\alpha, \quad t \in [-1, 1].$$

(hint: induction and chain rule)

Problem 2. Suppose $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is twice continuously differentiable.

- (a) Prove that the Taylor formula centered at $x_0 \in \mathbb{R}^n$ of order 2 is

$$f(x) = f(x_0) + (\nabla f(x_0))^\top (x - x_0) + \frac{1}{2} (x - x_0)^\top H_f(x_0) (x - x_0) + o(\|x - x_0\|^2).$$

- (b) Implement and plot the Taylor approximation of f of order 2 by a heatmap for the function $f : \mathbb{R}^2 \rightarrow \mathbb{R}$,

$$f(x, y) = x^4 + y^4 + x^2 + y^2.$$

Problem 3. Consider the function $f : \mathbb{R}^2 \rightarrow \mathbb{R}$,

$$f(x, y) = x^2 + y^2 + 2x + 4y.$$

- (a) Determine all local extrema of f .
(b) Plot f in Julia by a heatmap to check your calculations in Part (a).

Problem 4. Consider the function $f : \mathbb{R}^2 \rightarrow \mathbb{R}$,

$$f(x, y) = x^2 + 3y^2 + 2xy.$$

- (a) Determine all local extrema of f .
(b) Plot f in Julia by a heatmap to check your calculations in Part (a).